Lab 3 - Deliverable 2

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A. Rotor Configuration

Question A asks whether changing propellers 3 and 4 and reversing their rotation still allows independent position and heading control. Use the exercise numbering: in top view, x points right and y upward. Rotor positions are (-a,a), (-a,-a), (a,-a), (a,a), with a=d/sqrt(2). For u_i=omega_i squared, [f,tau_x,tau_y,tau_z]^T=Fu. This numbering is separate from the simulator numbering.

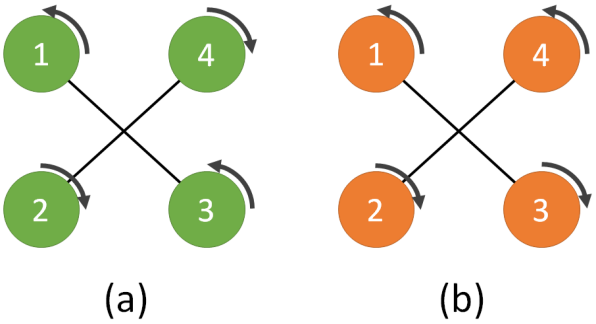


Figure A1. Exercise configurations a and b.

$$F_a = \begin{bmatrix} c_f & c_f & c_f & c_f \\ ac_f & -ac_f & -ac_f & ac_f \\ ac_f & ac_f & -ac_f & -ac_f \\ -c_d & c_d & -c_d & c_d \end{bmatrix} \qquad F_b = \begin{bmatrix} c_f & c_f & c_f & c_f \\ ac_f & -ac_f & -ac_f & ac_f \\ ac_f & ac_f & -ac_f & -ac_f \\ -c_d & c_d & c_d & -c_d \end{bmatrix}$$

Equation (A1). Allocation matrices for the two configurations.

Table A1. Configuration analysis for positive coefficients

Configuration	Result	Meaning
a	rank=4; |det|=16a^2c_f^3c_d	Four independent control components
b	rank=3; tau_z=-c_d tau_x/(a c_f)	Roll and yaw are coupled, not independently assignable

For positive a, c_f and c_d, the normal matrix has a nonzero determinant and rank 4. In configuration b, the first three rows are independent, but row 4 equals row 2 multiplied by -c_d/(a c_f), giving rank 3. Changing roll moment therefore forces a linked change in yaw moment. Position and yaw cannot be assigned independently in general. Special motions that satisfy this constraint may still be possible.

B. Drag Compensation

Question B asks how the given drag model could improve tracking. Estimate the drag first, then ask the rotors for an opposing force. R maps body vectors to the world frame, so compute body velocity as v_b=R^T v before using the model.

$$v^b = R^T v, \quad D = \mathrm{diag}(0.1, 0.1, 0.2), \quad F^b = -D\,(v^b \odot |v^b|)$$

Equation (B1)

D=diag(0.1,0.1,0.2). The product is component by component. Drag opposes the velocity and grows with its squared magnitude along each axis. With world z pointing upward, the force balance is:

$$m\dot{v} = -mge_3 + fRe_3 + RF^b$$

Equation (B2)

Set e_x=x-x_d and e_v=v-v_d. Transform the estimated drag to the world frame and subtract it from the baseline force. Subtracting the external force requests a force in the opposite direction:

$$A = -k_x e_x - k_v e_v + mge_3 + ma_d - R\widehat{F}^b$$

Equation (B3)

Construct the desired attitude from the compensated force and project it onto the current thrust axis:

$$b_{3d} = A/\|A\|, \quad f = A^T R e_3$$

Equation (B4)

If attitude tracking makes the thrust vector close to A, substitution into the force balance gives Equation (B5). A good estimate cancels most of the disturbance. Model error and noisy velocity leave a residual, so a flight comparison is still needed to establish any improvement.

$$m\ddot{e}_x \simeq -k_x e_x - k_v e_v + R(F^b - \widehat{F}^b)$$

Equation (B5)

Table B1. Derivation and proposed validation

Item	Scope in this work
Work completed	Derivation only; not added to the flight code or tested in a flight comparison
How to check it	Keep the path, gains and initial state fixed; compare errors and commands with compensation on and off

References: [1] MIT VNAV, Lab 3 Exercises (2023), vnav.mit.edu/labs_2023/lab3/exercises.html. [2] Lee, Leok and McClamroch, Geometric Tracking Control of a Quadrotor UAV on SE(3), IEEE CDC, 2010, pp. 5420-5425.